

Benjamin Eppinger

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Summary

Full-stack engineer with 7+ years shipping production software, most of it as the person who owns the problem end to end. Built Nova Backbone single-handedly after two failed off-the-shelf attempts: a 13-module operations platform (React/Node/PostgreSQL) that is now the company-wide system of record for global drone show operations, spanning aviation permitting, a ~35,000-unit fleet across 10 countries, crew scheduling, logistics, and travel, all behind a comprehensive role-based permission model. I've shipped mobile bi-directional event-driven telemetry for 300+ autonomous devices at Scythe and co-authored reusable patterns adopted by 1,000+ developers at Liberty Mutual. Ran an entire IT and security function solo, twice over. When off-the-shelf tools don't fit the problem, I build the ones that do.

Work Experience

Nova Sky Stories

Head of IT / Boulder, CO / Oct 2025 – Present

- Promoted to lead Corporate IT after eight months building the function from zero; now own strategy, budget, vendor relationships, and a 30+ SaaS portfolio serving a global workforce across three countries; established IT metrics and report quarterly to leadership.
- Architected and deployed Nova HelpHub, a company-wide intranet and IT ticketing layer integrating Microsoft, Google, Nova SSO Applications, Slack and more via SCIM behind automated provisioning scripts, eliminating manual account creation and cutting onboarding setup time by over 80%.
- Drove AI adoption strategy company-wide: ran a comprehensive AI tools audit, co-authored the company's first AI usage policy.
- Negotiated and managed vendor contracts including a 23-seat Maxon license (\$25k/year), 150-seat Slack/Notion/Google contracts; established Apple Developer and federated Microsoft 365 ecosystems from scratch.
- Led the Boulder warehouse and Berkeley California IT & security buildouts end to end: sourced ISP, designed network topology, deployed access control and badge systems, and coordinated vendor installation to deliver a fully operational facility on schedule and under budget.

Nova Sky Stories

Software Engineering & IT Operations Lead / Boulder, CO / Oct 2025 – Jun 2026

- Built and shipped Nova Backbone solo after two failed attempts at off-the-shelf tooling, first Notion, then a large ClickUp deployment, neither of which could accurately model how drone shows get planned. Shipped a proof of concept on my own initiative, won over production managers and leadership, and turned it into a funded multi-month project priority.
- Grew it into a 13-module operations platform on React, Node.js, PostgreSQL, and Railway, now serving as the company-wide system of record for 250+ show projects per year, 550+ aviation permits, a ~35,000-unit drone fleet across 10 countries, plus crew scheduling, travel, logistics, client management, and analytics.
- Selected and implemented Railway and Neon as the company's application and database platform, replacing an estate split between hundreds of loosely governed Google Cloud projects with opaque billing and a single DigitalOcean droplet running dozens of apps, databases, and environment. Replaced manual SSH deploys with Git-native delivery, UI-managed permissions, per-service cost attribution, and blast-radius isolation that removed a company-wide single point of failure.
- Built a synthetic data generator that seeds dev and staging with realistic, referentially consistent records across shows, permits, crew, travel, and fleet inventory, enabling full-scale testing and safe external demos without touching production data.
- Developed a milestone templating engine that generates 140+ tracked milestones per show project, including per-show-date execution milestones for multi-date runs, with operator-triggered regeneration and bulk reset.

- Built the notification layer, combining transactional email via Resend with an in-app notification center for record changes, upcoming actions, and slipped or overdue milestones. Routed date slips to affected parties for resolution instead of silently recomputing downstream dates, preserving manual scheduling decisions made for operational reasons.
- Designed a six-dimension health model per show, spanning production, permitting, design, scheduling, shipping, and overall status, with automatic rollups from underlying records and per-field manual override to surface at-risk shows on a triage dashboard.
- Implemented a multi-step approval workflow engine for aviation permitting, moving reviews from Expert to Manager to Routine Updates, with reviewer queues, overdue escalation, automated health scoring, and region-scoped views for EU and US operations.
- Designed hierarchical role-based access control spanning 18 roles and 14 permission groups, with section-level defaults, granular per-permission overrides, and a view-as-role mode for verifying access without maintaining test accounts.
- Built global fleet planning with time-series inventory across transit and repair pools in 10 international locations, visualized as stacked time-series charts and an interactive geospatial map with a date scrubber, backed by bulk CSV ingest with template generation and duplicate detection.
- Shipped a configurable data grid with user-defined saved views, column reordering and resizing, inline editing, and multi-field filtering, backed by PostgreSQL audit trails and rollback.
- Ran the entire IT function solo on the same timelines, and launched the revamped production website, resolving third-party integration issues including Osano cookie consent and Instagram embeds, and coordinating Cloudflare cache and DNS invalidation for a clean public launch.

BenTheITGuy LLC

Full-Stack Software Engineer (Contract) / Longmont, CO / Mar 2025 – Present

- Developed and delivered full-stack features for a gig-economy platform using React, TypeScript, and Node.js, collaborating directly with founders and product teams.
- Designed and deployed physical and network infrastructure using Ubiquiti UniFi, including access points, switching, gateway configuration, and running and terminating Ethernet for business environments.
- Provided consulting, web redevelopment, and technical deployment for a small recruiting firm, rebuilding its website, configuring domains, creating web content, and implementing outreach systems to support business growth.

Scythe Robotics

Staff Software Engineer, Web/Mobile & IT Operations / Longmont, CO / Jan 2023 – Jan 2025

- Architected and led a cross-platform React application to manage real-time fleet data, enabling sub-5-second updates for 300+ autonomous devices.
- Implemented MQTT for telemetry and GraphQL for mobile queries, optimizing each transport for real-time streaming versus request-response workflows.
- Designed device-agnostic OAuth and 2FA using Stytych, improving both security and user experience across enterprise applications.
- Delivered responsive UIs with bilingual support and offline-tolerant flows, improving usability for field teams in low-connectivity environments.
- Reduced hardware spend 66% through standardization and vendor negotiation, cutting costs while maintaining reliability.

Liberty Mutual Insurance

Software Developer, US Claims / Boston, MA / Jul 2019 – Jan 2023

- Built and maintained React and Node.js applications for high-volume claims processing, streamlining workflows for agents and customers.
- Implemented REST and GraphQL APIs on AWS Lambda, improving scalability and reducing operational overhead.
- Onboarded and mentored a newly formed claims development team, accelerating ramp-up and improving team velocity.

Liberty Mutual Insurance

Associate Software Developer, DNA Platform / Boston, MA / Jan 2020 – Jul 2021

- Developed reusable software code patterns for Node.js, React, Java, and AWS Lambda adopted by over 1,000 internal developers.
- Contributed to shared component libraries and design systems, ensuring consistency across multiple product teams.
- Contributed to a company-wide Kubernetes migration, improving scalability, deployment speed, and operational resilience.

Liberty Mutual Insurance

TechStart Software Developer & Team Lead / Boston, MA / Jul 2019 – Jan 2020

- Led a team to deliver a production texting channel and management hub, accelerating feature rollout and user support.
- Established coding standards and automated testing frameworks to improve code quality and deployment reliability.
- Designed a Bamboo CI/CD pipeline enhancement that reduced processing time by 10 minutes per commit.

Liberty Mutual Insurance

Software Development Intern / Boston, MA / May 2018 – Aug 2018

- Built a full-stack web application using Django, Jinja, and Python, earning 1st place in a company-wide software innovation challenge.

Education

Pennsylvania State University / University Park, PA / May 2019

M.S. Information Sciences & Technology (GPA 3.79) · B.S. Information Sciences & Technology, Design & Development (GPA 3.76) / GPA:

- Completed master's and bachelor's degrees concurrently in four years; specialized in UX design, human-computer interaction, and educational psychology.
- Resident Assistant, IST Special Living Option, led a 120-student technical residential community; organized tech-focused events, coding workshops, and cybersecurity challenges.

Skills

Languages & Frameworks

- TypeScript, JavaScript (ES6+), React (incl. Native & Next.js), Node.js, PostgreSQL, GraphQL, REST, Python, Django, Java, Angular, Vue, Ionic

APIs & Integrations

- REST/GraphQL API design, webhooks, MQTT, transactional emails (Resend, Mailgun, Sendgrid, Knock), third-party vendor integrations (SSO, mapping, consent, weather), OAuth 2.0 / 2FA (Stytch), API contract definition, event-driven architectures

Cloud & Infrastructure

- AWS (Lambda, S3, EC2), Railway, Neon, Docker, Kubernetes, IaC, CI/CD (GitHub Actions, GitLab, Bitbucket, Jenkins, Bamboo), Vercel, Heroku, Cloudflare, DigitalOcean, GCP

AI & Automation

- Claude Code, Cursor, OpenRouter, Hermes Agent, LLM-assisted development, workflow automation, AI policy authoring, AI tool evaluation, prompt engineering

Identity & Security

- SSO architecture, identity & access management, endpoint management, Google Workspace administration, SaaS governance, scoped API tokens, SSL/TLS management, security policy

Hobbies

- Indoor Rock Climbing / Volunteer Competition Climbing Judge / 3D Printing / Ice Skating / Gold Panning